Unit Exam 1 (Ch 20-22)_

- Due Oct 23 at 11:59pm
- Points 100
- Questions 50
- Available Oct 19 at 12am - Oct 23 at 11:59pm
- Time Limit 75 Minutes

Instructions

Exam 1 (Chapters 20 through 22)

You will have 75 minutes to answer 50 questions. Please make sure you have a quiet place with a good internet connection to take the exam. There are NO makeups for this exam, no exceptions.

You may use handwritten notes for the exam. You MUST show ALL notes to the camera before you begin your exam.

Your room scan must be sufficient to show a vertical and horizontal 360 view of your testing area, as well as your COMPLETE working area including the entire desk/table you are using. Once you begin the exam you must complete it in one sitting. If you leave the scanned testing area, the view of the camera, or exit out of the screen/exam your test will automatically be submitted or be considered submitted at that time. Please reach out to Professor Johnson with any questions BEFORE you begin the exam.

Make sure to review all Exam Rules. Failure to follow ANY exam rules will result in a grade deduction and possibly a zero.

Good Luck!

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	31 minutes	100 out of 100

ⓘ Correct answers will be available Oct 23 at 12am - Oct 26 at 12am.

Score for this quiz: 100 out of 100
Submitted Oct 20 at 7:27pm
This attempt took 31 minutes.

Question 1

2 / 2 pts

Why must a virus enter a living cell to reproduce?

- Its capsid performs photosynthesis only inside cells.
- It must obtain a nucleus from another virus.
- It depends on the host cell's machinery and energy.
- It produces offspring by binary fission only inside cells.

Question 2

2 / 2 pts

What is the primary structural role of a viral capsid?

- It forms a protein covering around the viral genome.
- It makes ATP for the virus.
- It converts viral RNA into host ribosomes.
- It functions as the virus's nucleus.

Question 3

2 / 2 pts

A virus attaches to one type of host cell but cannot attach to another. Which difference best explains this specificity?

- The cells have different surface receptors.
- The cells contain different amounts of water.
- Only the susceptible cell has a plasma membrane.
- Only the susceptible cell contains DNA.

Question 4

2 / 2 pts

Which event distinguishes a lysogenic bacteriophage infection from an immediately lytic infection?

- Viral DNA integrates into bacterial DNA and is copied with it.
- The bacterium immediately bursts and releases new viruses.
- Viral proteins replace all bacterial DNA before attachment.
- The phage reproduces independently outside the bacterium.

Question 5

2 / 2 pts

A drug blocks reverse transcriptase in HIV. Which process is directly inhibited?

- Translation of viral proteins by the capsid.
- Formation of bacterial peptidoglycan.

- Synthesis of RNA using a protein as a template.
- Synthesis of DNA using viral RNA as a template.

Question 6

2 / 2 pts

An animal virus leaves a cell by budding. What can it acquire during this process?

- A bacterial chromosome.
- A membranous envelope.
- A functional mitochondrion.
- A peptidoglycan wall.

Question 7

2 / 2 pts

Which pairing correctly identifies a viroid and a prion?

- A viroid is infectious DNA; a prion is infectious RNA.
- A viroid is a bacterial cell; a prion is a fungal cell.
- A viroid is a protein capsid; a prion is a viral envelope.
- A viroid is infectious RNA; a prion is an infectious protein.

Question 8

2 / 2 pts

A student finds a small cell whose chromosome lies in a nucleoid and which lacks membrane-bound organelles. How should the cell be classified?

- As a prokaryote.
- As an animal cell.
- As a fungus.
- As a protist.

Question 9

2 / 2 pts

Which process directly produces new bacterial cells during ordinary asexual reproduction?

- Transduction.
- Binary fission.
- Conjugation.
- Transformation.

Question 10

2 / 2 pts

A bacterium takes up free DNA released by a dead bacterium and incorporates it into its genome. Which process occurred?

- Transduction.
- Conjugation.
- Binary fission.
- Transformation.

Question 11

2 / 2 pts

An antibiotic resistance gene passes from a donor bacterium to a recipient through a conjugation pilus. Which process occurred?

- Transformation.
- Conjugation.
- Meiosis.
- Transduction.

Question 12

2 / 2 pts

A bacteriophage carries bacterial DNA from one bacterial cell to another. This is an example of which process?

- Transduction.
- Transformation.
- Conjugation.
- Budding.

Question 13

2 / 2 pts

After a correctly performed Gram stain, a bacterial sample appears purple. Which interpretation is most appropriate?

- The bacteria are Gram-positive and retained the crystal violet dye complex.
- The bacteria are Gram-negative and retained only the counterstain.
- The bacteria lack a cell wall.
- The bacteria must belong to Archaea.

Question 14

2 / 2 pts

A bacterial culture grows both in the presence and absence of oxygen. Which description best fits it?

- Facultative anaerobe.

- Obligate aerobe.
- Obligate intracellular virus.
- Obligate anaerobe.

Question 15

2 / 2 pts

What is the main benefit of endospore formation to a bacterium?

- Transfer of DNA through a pilus.
- Capture of light for photosynthesis.
- Survival during unfavorable conditions.
- Rapid production of many daughter cells.

Question 16

2 / 2 pts

Which feature helps distinguish Archaea from Bacteria?

- Archaea have membrane-bound nuclei.
- Archaea lack genetic material.
- Archaea reproduce only by meiosis.
- Archaeal cell walls lack peptidoglycan.

Question 17

2 / 2 pts

An archaeon requires high salt concentrations for growth. Which group best describes it?

- Thermoacidophiles.
- Halophiles.
- Cyanobacteria.
- Methanogens.

Question 18

2 / 2 pts

Which contribution of cyanobacteria had a major effect on Earth’s early atmosphere?

- Conversion of oxygen into peptidoglycan.
- Production of methane in animal intestines.
- Release of oxygen through photosynthesis.
- Removal of all atmospheric nitrogen.

Question 19

2 / 2 pts

According to the endosymbiotic hypothesis, chloroplasts originated from which organisms?

- Methanogenic archaea.
- Bacteriophages.
- Parasitic fungi.
- Cyanobacteria.

Question 20

2 / 2 pts

A protist photosynthesizes in light but also ingests food particles. Which term describes its nutrition?

- Exclusively photoautotrophic.
- Obligately parasitic.
- Mixotrophic.
- Exclusively saprotrophic.

Question 21

2 / 2 pts

A reduction in photosynthetic plankton would most directly reduce which ecosystem functions?

- Primary production and oxygen release.
- Conversion of fungal hyphae into roots.
- Production of methane by all protists.
- Formation of peptidoglycan in animals.

Question 22

2 / 2 pts

A hiker develops diarrhea after drinking untreated stream water containing Giardia cysts. What role did the cysts serve?

- They carried out conjugation between bacteria.
- They produced fungal basidiospores.
- They converted the parasite into a virus.
- They allowed the parasite to survive and transfer to a new host.

Question 23

2 / 2 pts

Which structure helps Euglena orient toward light for photosynthesis?

- A silica test.
- An eyespot associated with light detection.
- A fungal ascus.
- A conjugation pilus.

Question 24

2 / 2 pts

Which organism causes African sleeping sickness and is transmitted by the tsetse fly?

- Entamoeba histolytica.
- Trypanosoma brucei.
- Plasmodium vivax.
- Giardia lamblia.

Question 25

2 / 2 pts

Coral-associated dinoflagellates provide organic nutrients to their hosts. What do the corals provide in return?

- Shelter, nutrients, and protection.
- Peptidoglycan and bacterial chromosomes.
- Asci and basidiospores.
- A supply of red blood cells.

Question 26

2 / 2 pts

A coastal dinoflagellate bloom causes fish deaths and makes shellfish dangerous to eat. Which mechanism best explains these effects?

- Formation of fungal mycelia inside every fish.
- Production of powerful toxins by certain dinoflagellates.
- Production of bacterial endospores by shellfish.
- Conversion of seawater into freshwater.

Question 27

2 / 2 pts

Which pairing correctly matches a protist with its disease and vector?

- Entamoeba histolytica — sleeping sickness — tsetse fly.
- Trypanosoma brucei — amoebic dysentery — mosquito.
- Plasmodium — malaria — mosquito.
- Giardia — malaria — tsetse fly.

Question 28

2 / 2 pts

In a ciliate, which nuclear structure primarily controls normal cell metabolism?

- Nucleoid.
- Macronucleus.
- Apicoplast.
- Micronucleus.

Question 29

2 / 2 pts

A plankton sample contains single-celled organisms with ornate silica shells. Which identification is most likely?

- Choanoflagellates.
- Cellular slime molds.
- Diatoms.
- Brown algae.

Question 30

2 / 2 pts

Which feature distinguishes oomycetes from true fungi as described in the slides?

- Their cell walls are composed largely of cellulose.
- They obtain all nutrients by photosynthesis.
- They lack all reproductive spores.
- Their cells are prokaryotic.

Question 31

2 / 2 pts

A fossil has a multichambered calcium carbonate test with openings for threadlike pseudopods. Which group does it represent?

- Ciliates.
- Dinoflagellates.
- Foraminiferans.
- Diatoms.

Question 32

2 / 2 pts

Which algal group is identified as the source of agar and carrageenan?

- Red algae.
- Brown algae.
- Golden brown algae.
- Diatoms.

Question 33

2 / 2 pts

Which group of green algae is identified in the slides as the lineage from which land plants arose?

- Oomycetes.
- Charophytes.
- Radiolarians.
- Dinoflagellates.

Question 34

2 / 2 pts

During food shortage, individual amoeboid cells aggregate into a pseudoplasmodium. Which organisms show this behavior?

- Red algae.
- Cellular slime molds.
- Plasmodial slime molds.
- Diatoms.

Question 35

2 / 2 pts

In the alternation of generations life cycle of Ulva, which process produces haploid spores from the diploid sporophyte?

- Binary fission.
- Fertilization.
- Meiosis.
- Budding.

Question 36

2 / 2 pts

How do fungi typically obtain nutrients from dead organic matter?

- They ingest food into a stomach cavity.
- They secrete digestive enzymes and absorb the resulting molecules.
- They use chloroplasts to produce all required nutrients.
- They draw nutrients through a conjugation pilus.

Question 37

2 / 2 pts

How does a network of thin hyphae benefit a fungus?

- [x] It provides a large surface area for nutrient absorption.
- [ ] It prevents all contact with the food source.
- [ ] It enables photosynthesis without pigments.
- [ ] It replaces the need for a plasma membrane.

Question 38

2 / 2 pts

Which pairing correctly describes fungal cell composition and food storage?

- [x] Chitin cell walls and glycogen storage.
- [ ] Silica cell walls and laminarin storage.
- [ ] Peptidoglycan cell walls and glycogen storage.
- [ ] Cellulose cell walls and starch storage.

Question 39

2 / 2 pts

How do aseptate hyphae differ from septate hyphae?

- [ ] Aseptate hyphae are always photosynthetic.
- [ ] Aseptate hyphae contain no nuclei.
- [ ] Aseptate hyphae have peptidoglycan walls.
- [x] Aseptate hyphae lack cross walls that partition them into cells.

Question 40

2 / 2 pts

A fungal hyphal compartment contains two separate haploid nuclei from different mating types. Which term describes it?

- [ ] Diploid with a single nucleus.
- [ ] Prokaryotic.
- [ ] Anucleate.
- [x] Dikaryotic.

Question 41

2 / 2 pts

After nuclear fusion produces a diploid nucleus in fungal sexual reproduction, which process restores haploid nuclei?

- [ ] Binary fission.
- [x] Meiosis.
- [ ] Budding.
- [ ] Conjugation pilus formation.

Question 42

2 / 2 pts

A yeast cell develops a small outgrowth that enlarges and separates. Which reproductive process is occurring?

- Conjugation.
- Spore fusion.
- Budding.
- Transduction.

Question 43

2 / 2 pts

What distinguishes a zoospore from the usual nonmotile spores of terrestrial fungi?

- A zoospore lacks genetic material.
- A zoospore is always a multicellular embryo.
- A zoospore has a flagellum and can swim.
- A zoospore is produced only by bacteria.

Question 44

2 / 2 pts

Anaerobic fungi in an herbivore’s gut help the animal obtain energy from plant material. Which fungal activity supports this benefit?

- Enzymatic breakdown of plant structural materials.
- Conversion of animal cells into spores.
- Production of peptidoglycan for the animal.
- Photosynthetic production of glucose in chloroplasts.

Question 45

2 / 2 pts

A frog population declines after a fungal infection damages the animals’ skin. Which fungus is associated with this problem in the slides?

- Batrachochytrium dendrobatidis.
- Rhizopus stolonifer.
- Penicillium used in cheese production.
- Saccharomyces cerevisiae.

Question 46

2 / 2 pts

Which specialized Rhizopus hyphae grow into bread, anchor the fungus, and carry out digestion?

- Basidia.
- Sporangiophores.
- Rhizoids.
- Stolons.

Question 47

2 / 2 pts

Where are the sexual spores of sac fungi produced?

- On basidia associated with a basidiocarp.
- Inside bacterial endospores.
- Inside asci associated with an ascocarp.
- Within a ciliate macronucleus.

Question 48

2 / 2 pts

A gilled mushroom releases sexual spores from club-shaped structures on its gills. What are these structures called?

- Rhizoids.
- Asci.
- Basidia.
- Heterocysts.

Question 49

2 / 2 pts

Which association forms a lichen?

- A fungus associated with a photosynthetic alga or cyanobacterium.
- A bacteriophage integrated into a bacterial chromosome.
- A fungus associated exclusively with an animal root.
- Two viruses sharing a protein capsid.

Question 50

2 / 2 pts

Plants with mycorrhizae grow better in phosphate-poor soil than plants without them. Which exchange best explains this mutualism?

- Fungi improve mineral uptake; plants supply carbohydrates.
- Plants digest fungi; fungi supply sunlight.
- Fungi supply chloroplasts; plants supply bacterial endospores.

- Plants supply peptidoglycan; fungi supply viral DNA.

Quiz Score: 100 out of 100